

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2017

Subject: 392152 Chemistry II

Section: 15-18

Date: 7 March 2018

Time: 13.00-15.00

Name: _____ ID: _____ Section: _____

Instructions:

1. The examination has 9 pages (including this page), 2 sections and a total score of 48 points.
2. Write all your solutions and answer on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.

Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.

Relative atomic mass of H =1, He =4, C =12, N =14, O =16, F =19, Na =23, S =32, Cl =35.5,
Ar =40, K =39

Gas constant; $R = 0.082 \text{ atm.L.mol}^{-1}.\text{K}^{-1}$ or $8.314 \text{ J.mol}^{-1}.\text{K}^{-1}$

	Multiple Choice	Part 2.1	Part 2.2	Part 2.3	Part 2.4	Total
score						

Part 1. Multiple Choice (16 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				

Question no.	a.	b.	c.	d.
9				
10				
11				
12				
13				
14				
15				
16				

- The particles in a liquid are usually
 - closer together and lower in energy than those in a solid.
 - farther apart and higher in energy than those in a gas.
 - closer together and lower in energy than those in a gas.
 - farther apart and lower in energy than those in a solid.
- At pressures greater than 1 atm, water will boil at
 - a temperature higher than 100°C .
 - 100°C
 - a temperature lower than 100°C .
 - 40°C

3. Amorphous solids have solid state properties but they do not possess
- a. specific volume
 - b. virtual shape maintenance
 - c. orderly arrangement
 - d. none
4. Water boils at a lower temperature on Doi Inthanon than it does at sea level because
- a. the pressure is lower on Doi Inthanon than it is at sea level.
 - b. the pressure is higher on Doi Inthanon than it is at sea level.
 - c. the heat source is not as hot at higher altitudes.
 - d. the heat source is hotter at higher altitudes, so water boils at lower temperatures.
5. When a gas changes to solid, it is known as that is an process.
- a. condensation exothermic
 - b. condensation endothermic
 - c. deposition endothermic
 - d. deposition exothermic
6. What are standard temperature and pressure conditions for gases?
- a. 0°C and 1 Torr
 - b. 0 K and 760 mmHg
 - c. 273 K and 101.3 kPa
 - d. 273°C and 1 atm
7. If the volume of gas is decreased and pressure is constant, the temperature will
- a. increase
 - b. decrease
 - c. remain the same
 - d. increase double
8. What is Gay-Lussac's law?
- a. The pressure is directly proportional to the absolute temperature at a constant volume.
 - b. The pressure is directly proportional the volume of gas at a constant temperature.
 - c. The volume of a gas is directly proportional to the absolute temperature at a constant pressure.
 - d. The volume of a gas is directly proportional to the number of mole of gas at a constant pressure.

9. Three containers of equal volume are filled with equal masses of helium (He), nitrogen (N_2) and carbon dioxide (CO_2) gas respectively. The gases in the three containers are all at the same temperature. Which one of the following statements is correct regarding the pressure of the gases?

- a. All three gases will be at the same pressure
- b. The helium will be at the greatest pressure
- c. The nitrogen will be at the greatest pressure
- d. The carbon dioxide will be at the greatest pressure

10. Balloon is filled by helium gas until its volume doubled at constant temperature and pressure. Which one of the following statements is correct?

- a. Number of molecule is constant
- b. Number of molecule increases 2 times
- c. Number of molecule decreases 2 times
- d. None of it

11. Which statement is CORRECT?

- a. Effusion is a gradual movement of molecule according to its kinetic properties
- b. Diffusion is a movement of molecule through the tiny pore into another gaseous region
- c. Lighter gas molecules effuse faster than the heavier molecule
- d. a. b. and c. are correct

12. Which statement is CORRECT?

- a. Real gas behave like ideal gas except at low pressure and low temperature
- b. Absolute gas exhibit a strong interaction between molecule
- c. Behavior of ideal gas follows Boyle's law
- d. b. and c. are correct

13. What is NOT one of the solution properties?

- a. clear and transparent
- b. homogeneous
- c. no light reflection
- d. size of solute particles 10 - 10,000 Angstroms

14. Which statement is CORRECT?

- a. Light cannot pass through solution
- b. Light is scattered by colloidal particles
- c. Tyndall effect cannot be found in headlight in the fog
- d. Invisible particles are detected in suspension

15. Which solution does contain the highest Calcium carbonate (CaCO_3) quantitatively?

- a. 100 mL of CaCO_3 1 Molar
- b. 250 mL of CaCO_3 0.5 Molarity
- c. 200 mL of CaCO_3 5% (w/v)
- d. 250 mL of CaCO_3 2% (w/w)

16. How many moles of NaCl are there in 10 mL of a solution with a concentration of 0.5 M?

- a. 0.005 mol
- b. 0.05 mol
- c. 0.5 mol
- d. 1 mol

Part 2 Write your answer in the space provided for each question (32 points)

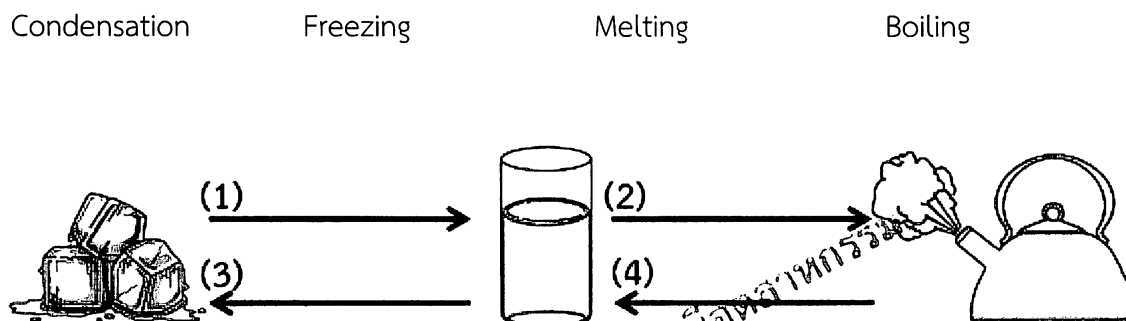
1.1 Give a characteristic of the following states

Gas (0.5 pt) :

Liquid (0.5 pt):

Solid (0.5 pt) :

1.2 Use technical terms and the figure as shown below and write correct technical terms to match process names (1)-(4) for the state changes of water on top of arrows (2 pts)



Evaporation

Deposition

Sublimation

Crystallization

1.3 Explain water vaporization (4 pt) Hint: describe the process in term of intermolecular force

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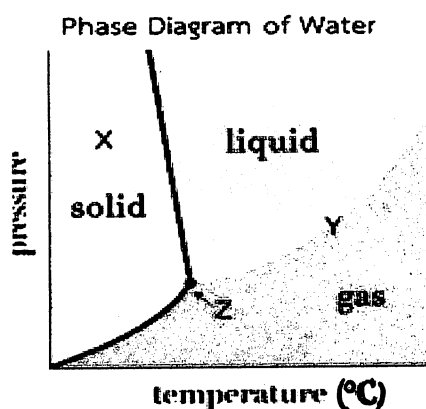
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1.4 Define an amorphous solid and a crystalline solid (2 pt)

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1.5 Specify state (s) in which X, Y, and Z are located in the phase diagram of water as shown below



X (0.5 pt):

Y (0.5 pt):

Z (0.5 pt):

2.1 Calculate the initial volume of a sample of gas at 1.50 atm if its volume is changed to 80 mL as its pressure is changed to 1,520 mmHg at constant temperature (2 pts.)

2.2 Calculate the initial volume of gas that its absolute temperature tripled and its final volume is 6 L (2 pts.)

2.3 The gas in an aerosol can is under a pressure of 5 atm at a temperature of 25 °C. It is dangerous to dispose of an aerosol can by incineration. What would the pressure in the aerosol can be at a temperature of 845 °C (2 pts.)

2.4 Calculate the final pressure of a gas that is expanded from 800 mL at 25 °C and 1 atm to 2 L at 35 °C (2pts.)

3.1 Arrange these gases (CO_2 , Cl_2 , HF , SO_3) in order from lowest to highest rate of effusion. (2pts.)

3.2 An unknown gas effuses at a rate one-third as fast as that of Helium gas (He). What is the molar mass of the unknown gas? (2pts.)

3.3 What is the density of nitrogen monoxide (NO) at 1,520 Torr and 27°C ? (2pts.)

3.4 A sample of Argon gas (Ar) has a volume of 500 cm^3 and a pressure of 202.6 kPa at 127°C . What is the mass of Argon gas in grams? (2pts.)

4. Solution, Concentration (8 pts)

4.1 Complete the following table.

(4 pts)

Concentration of a solution	Amount of the solute	Amount of the solvent	Amount of the solution
2.0 mol/kg solvent	0.5 mol		-- do not fill in --
..... mol/L	90 g $C_6H_{12}O_6$	-- do not fill in --	2000 mL
10 %w/w		45 g.	-- do not fill in --
mole fraction of solute = 0.15		-- do not fill in --	50 mol

Write your solution

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4.2 Calculate the following solution.(4 pts.)

- the mass in gram of solute needed to be prepare 100 mL of a 2.5 M KCl solution
- the molality of the solution prepared by dissolving 53.5 g of NH_4Cl in 3 kg of water

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